Quantitative Online Survey of RItrainPlus

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The quantitative study is dedicated to the expectations of all those who work in European RIs and CFs. The aim of the survey is to define an understanding about what makes these institutions a success and what is important to their members with diverse backgrounds, tasks, interests and needs: Who are they and what is it like to work in a European RI or CF? What challenges and demands do those who support the first pillar of scientific excellence encounter day after day? Which skills do they need to fulfil their tasks? What type of training or support do they need to make it work? The quantitative questionnaire deals with these questions and helps to create the structure and contents of the future curricula.

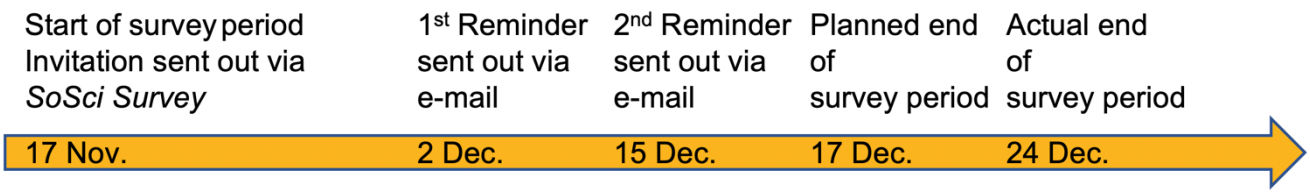

Figure 1 – Timeline of the RI:TRAIN Plus Survey.

Aims and Scope – Planning and Field Period

Accordingly, a quantitative online survey was fielded between November 17 and December 24, 2021, resulting in a field period of 37 days. The field phase was originally planned to last for 30 days, however, based on feedback during this time, we extended the time provided to complete the survey. The target population was defined as employees who work in European RI, particularly the ESFRI landmarks and projects, and in the CFs connected. The complete list contains 493 entries and can be found in the ANNEX. Similar to within the RI:TRAIN Plus project group, it was agreed upon that not only top level managers who lead European or national initiatives but also sites (e.g. subnational or national nodes) should be addressed. The questionnaire had to be inclusive for a wide range of individuals coming from different cultural and national backgrounds, scientific disciplines, and especially institutions that were made up by heterogenous and often highly specific structures.

A multi-step procedure was used to contact the individuals. Firstly, all the landmarks defined were contacted on November 17 via e-mail, sent from the online-survey platform *SoSci Survey*; two reminder waves, distributed directly via e-mail sent from the personal account of a researcher involved followed between December 2 and 15 (in accordance with 'individual out of office' notes). Christmas greetings were sent to all contacts on December 20, together with a 'thank you' for their active participation and a final reminder that the questionnaire would be online until December 24. Additionally, the co-ordinators of RI:TRAIN Plus spread the information that the survey was online (see Figure 2). The population could be defined and strategically contacted via both of these. However, an exact number of individuals in this population cannot be provided because the specific number of members and staff in individual RIs and their representatives in particular regions and countries were hard to define. This situation has to be highlighted, as it makes it evident that an adequate sampling frame for the whole population was missing and, consequently, completing a full probability sample was not possible. While this limits the potential for drawing conclusions based on interference via model building (see Cornesse et al., 2020), it enables us to research a otherwise hard to access population and research in other fields shows that the data quality – if controlled and argued for – is sound enough for descriptive and strictly theory-based models (Quatember, 2019; Prandner & Weichbold, 2019). These limitations have to be considered and results have to be read accordingly.

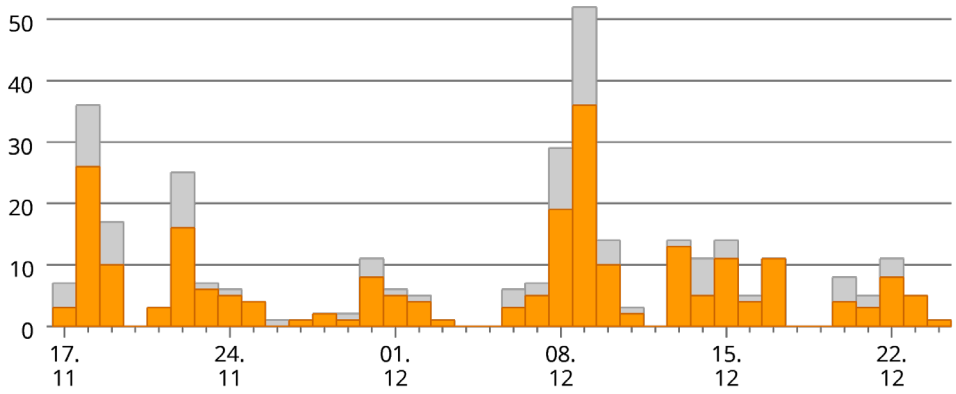

Figure 2 – Response to the survey during the field phase (orange bars: complete datasets, grey bars: additional useable datasets).

However, the approach made it possible to track and cover those who accessed the questionnaire, the completion rate and potential methodological issues. There have been no issues that could be identified. The questionnaire was accessed 1584 times over the course of the field period and 330 useable datasets were generated. Accordingly, we have an efficient translation rate of interview attempt to interview of 21 %. Of those 330, more than 70 % (235) had no missing values; the rest of the participants skipped some questions, but provided enough data to be included in the analysis.

These values can be seen as very good for the mode of an open online survey that targets a highly specialised group, without any incentives or other motivators. However, it has to be stressed that self-selection may be a potential issue, as it is for all kinds of online surveys. To account for this, constant data and plausibility were conducted during and after the field phase. Those showed some with misleading information (e.g. the year of birth was stated to be in the 2070s); such cases very evaluated in a more comprehensive way. If cases were deemed unfit for use after this evaluation (e.g. a simple typo could be ruled out), the respondent was deleted from the dataset.

Technical Set-up, Access to Data and GDPR Compliance

As highlighted previously, the survey was conducted via an online questionnaire hosted by the German online survey provider *SoSci Survey (www.soscisurvey.de)*. The server hosts the data for a period of six months after completion of the data collection and is physically set up in Munich, Germany. Access to the survey platform and the raw data collected was only provided to three researchers involved: Dimitri Prandner and Philip Sinner, the principal researchers, and Katrin Hasengruber, who helped with technical issues and programming. Furthermore, the raw dataset was made available to AUSSDA – the Austrian Social Science Data Archive – data manager Nikolai Soran. All of those individuals worked at the Johannes Kepler University (JKU) at the time of data collection and processing and processed the data in compliance with EU General Data Protection Regulations (GDPR). In addition to the server of *SoSci Survey*, the dataset is only saved on local hard drives and JKU servers.

The data are highly sensitive as they contain information on country of residence and origin, age, sex, career-track and affiliation to specific RIs and CFs, and the raw dataset had to be converted into a scientific use file that matches GDPR regulations. This dataset will subsequently be available via AUSSDA in long-term storage formats, useable with open source and commercial statistics software.

Most of the data is collected in a numeric format, while some fill-in questions contain strings. All data processing is handled with IBM SPSS 27 or 28.

Development of the Questionnaire and Modules

The questionnaire was developed in a multistage process, as the methodological and structural limitations became known. After an initial meetings in July with the WP2 team of RI:TRAIN Plus, several smaller meetings with a core group to exchange ideas and concepts for the questionnaire were shared. As highlighted in the first section, one of the main issues was tied to the fact that the project aims to address a highly heterogeneous community, which comes from service providers that are far removed from scientific practices, at the one end, and highly competitive scientists at the top of their corresponding disciplines, at the other. Furthermore, language barriers had to be considered and specific vocabulary used in different organisational and national contexts (Dillman et al., 2014, p. 463). Such problems often stress the reliability and validity of results when not considered adequately (for a starting point of the discussion concerning the associated problems, see Verba, 1973)

This necessitated structuring the questionnaire in such a way that it was both accessible and in depth enough to understand the context of each and every respondent (see Creswell & Creswell, 2018). Consequently, the decision was made to also cover structural information about the work context of the researcher population and how they perceive their situation, as it was expected that the circumstances experienced may not match available data on the set up of RIs and CFs.

Accordingly, we were interested in not only demographic data but also the professional socialisation of the individuals (studies, stays abroad, changes of positions) and their current situation (type of contract, number of RI or CF employees at the site, potential leadership role, amount of RI- or CF-relevant work in comparison to the general workload).

Additionally, several modules from surveys coming from studies in the field of the sociology of professions were used to get a general understanding of the tasks that both managers and staff have to do, their wishes if they want to do less or more of them, and more specific questions on the role of their RI or CF in the context of scientific work. This was seen mostly in the context of which services they provide for whom, what kind of materials or data they handle as well as their scope. Especially previous research done by the members of RI:TRAIN Plus proved to be valuable here in order to contextualise the situation. This was further contextualised with questions on satisfaction, work environment, career drivers and professional roles, which were meant to give insights into what kinds of skills or backgrounds led individuals to progress and succeed in the field.

Driven by the literature and previous studies, these questions were the foundation for the questionnaire, that was in the next step pretested by the RI:TRAIN Plus WP2 team and discussed. The first round of feedback identified that the questionnaire was too long to meet the targeted 15-minute survey time and was adjusted accordingly. In order to reduce the load on the participants, it was decided to cut parts of the more in-depth socialisation questions and questions on professional roles. In a second round of pretests, the staff of AUSSDA and personnel from medical and life sciences were included to do both a functional and content-based pretest. Several items were added to the questionnaire during this round that address the situation in the medical field more comprehensively and the wording was adjusted to be more inclusive for people outside of the academic context.

A final round of pretests was done subsequently, again with the group of WP2, and the questionnaire was finalised. A total of 56 pretests were conducted by 33 individuals, 16 of whom did more than one round of pretesting. However, there were specific remarks from the Consortium of European Science Data Archives (CESSDA) that the questionnaire may not address all of their members. While the project team discussed those issues both with the team of WP2 and CESSDA, certain aspects could only be acknowledged, as the aim was to generate an inclusive questionnaire that could cover a large number of RIs and CFs. This limitation has to be considered when reading the results and planning further projects that may focus in more depth on specific disciplines or institutions and may have different requirements. However, we used the tailored design method to create a broad survey that is able to reflect the whole situation at hand (Dillman et al., 2014; Creswell & Creswell, 2018)

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